

MTL106: Probability and Stochastic Processes

(Lecture 14)

February 3, 2026

Functions of Several Random Variables: Let $X_1, X_2, \dots, X_n$ be random variables defined on a probability space $(\Omega, \mathcal{F}, P)$. Often, we deal with functions of $X_1, X_2, \dots, X_n$ such as $X_1 + X_2$, $X_1 - X_2$, $X_1 X_2$, $\min(X_1, X_2, \dots, X_n)$, and so on. Are these also random variables? If so, how do we compute their distribution given the joint distribution of $X_1, X_2, \dots, X_n$? What functions of $(X_1, X_2, \dots, X_n)$ are random variables?

Theorem 1. Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a Borel-measurable function, i.e., if $B \in \mathcal{B}^m$, then $g^{-1}(B) \in \mathcal{B}^n$. If $\mathbf{X} = (X_1, X_2, \dots, X_n)$ is an n -dimensional random variable ($n \geq 1$), then $g(\mathbf{X})$ is an m -dimensional random variable.

Proof. For $B \in \mathcal{B}^m$,

$$\{g(X_1, X_2, \dots, X_n) \in B\} = \{(X_1, X_2, \dots, X_n) \in g^{-1}(B)\},$$

and since $g^{-1}(B) \in \mathcal{B}^n$, it follows that $\{(X_1, X_2, \dots, X_n) \in g^{-1}(B)\} \in \mathcal{F}$, which concludes the proof.

In particular, if $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a continuous function, then $g(X_1, X_2, \dots, X_n)$ is a random variable.

Computation of the Distribution of $g(X_1, X_2, \dots, X_n)$: Suppose $Y = g(X_1, X_2, \dots, X_n)$ is real-valued, and let $y \in \mathbb{R}$. Then

$$\begin{aligned} P\{Y \leq y\} &= P\{g(X_1, X_2, \dots, X_n) \leq y\} \\ &= \begin{cases} \sum_{\{(x_1, \dots, x_n) : g(x_1, \dots, x_n) \leq y\}} P(X_1 = x_1, \dots, X_n = x_n), & \text{discrete case,} \\ \int_{\{(x_1, \dots, x_n) : g(x_1, \dots, x_n) \leq y\}} f(x_1, \dots, x_n) dx_1 \cdots dx_n, & \text{continuous case.} \end{cases} \end{aligned}$$

Note that in the continuous case we can obtain the PDF of $Y = g(X_1, X_2, \dots, X_n)$ by differentiating the CDF, $P\{Y \leq y\}$ with respect to y , provided that Y is also of the continuous type. In the discrete case it is easier to compute $P\{g(X_1, X_2, \dots, X_n) = y\}$.

Example 1. Consider the following bivariate PMF:

$$P\{X = x, Y = y\} = \frac{(x + y + k - 1)!}{x! y! (k - 1)!} p_1^x p_2^y (1 - p_1 - p_2)^k,$$

where $x, y = 0, 1, 2, \dots$, $k \geq 1$ is an integer, $p_1, p_2 \in (0, 1)$, and $p_1 + p_2 < 1$.

Obtain the PMF of $X + Y$.

Solution. Let $U = X + Y$ and $V = Y$. Then $u = x + y$, $v = y$ defines a one-to-one mapping from

$$A = \{(x, y) : x, y = 0, 1, 2, \dots\}$$

onto

$$B = \{(u, v) : v = 0, 1, 2, \dots, u; u = 0, 1, 2, \dots\}.$$

The inverse mapping is $x = u - v$, $y = v$.

Thus, the joint PMF of (U, V) is

$$P\{U = u, V = v\} = \frac{(u + k - 1)!}{(u - v)! v! (k - 1)!} p_1^{u-v} p_2^v (1 - p_1 - p_2)^k, \quad (u, v) \in B.$$

The marginal PMF of U is

$$\begin{aligned} P\{U = u\} &= \frac{(u + k - 1)!(1 - p_1 - p_2)^k}{(k - 1)! u!} \sum_{v=0}^u \binom{u}{v} p_1^{u-v} p_2^v \\ &= \frac{(u + k - 1)!(1 - p_1 - p_2)^k}{(k - 1)! u!} (p_1 + p_2)^u = \binom{u + k - 1}{u} (p_1 + p_2)^u (1 - p_1 - p_2)^k, \quad u = 0, 1, 2, \dots \end{aligned}$$

Example 2. Consider the bivariate PDF

$$f(x_1, x_2) = \begin{cases} 2, & 0 \leq x_1 \leq x_2 \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Obtain the PDF of $X_1 + X_2$.

Solution. Let $Y = X_1 + X_2$. For $y < 0$, $P(Y \leq y) = 0$, and for $y > 2$, $P(Y \leq y) = 1$.

For $0 \leq y \leq 2$,

$$P(Y \leq y) = \iint_{\{0 \leq x_1 \leq x_2 \leq 1, x_1 + x_2 \leq y\}} 2 \, dx_1 dx_2.$$

Case I: $0 \leq y \leq 1$.

$$P(Y \leq y) = \int_0^{y/2} \left(\int_{x_1}^{y-x_1} 2 \, dx_2 \right) dx_1 = 2 \int_0^{y/2} (y - 2x_1) \, dx_1 = \frac{y^2}{2}.$$

Case II: $1 \leq y \leq 2$.

$$P(Y \leq y) = 1 - P(Y > y) = 1 - \int_{y/2}^1 \left(\int_{y-x_2}^{x_2} 2 \, dx_1 \right) dx_2 = 1 - 2 \int_{y/2}^1 (2x_2 - y) \, dx_2 = 1 - \frac{(y - 2)^2}{2}.$$

Hence, the PDF of Y is

$$f_Y(y) = \begin{cases} y, & 0 \leq y \leq 1, \\ 2 - y, & 1 \leq y \leq 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Computation of the Distribution of $g(X_1, X_2, \dots, X_n)$ Using Jacobian of Transformation Method (Continuous Case): Let $(X_1, X_2, \dots, X_n)$ be an n -dimensional continuous random variable with PDF $f(x_1, x_2, \dots, x_n)$.

1. Let

$$y_1 = g_1(x_1, x_2, \dots, x_n), \quad y_2 = g_2(x_1, x_2, \dots, x_n), \quad \dots, \quad y_n = g_n(x_1, x_2, \dots, x_n)$$

be a one-to-one mapping of $\mathbb{R}^n$ into itself.

2. Assume that the inverse transformation

$$x_1 = h_1(y_1, \dots, y_n), \quad \dots, \quad x_n = h_n(y_1, \dots, y_n)$$

exists and is continuous.

3. Assume that the partial derivatives

$$\frac{\partial x_i}{\partial y_j}, \quad 1 \leq i, j \leq n,$$

exist and are continuous.

4. Assume that determinant of the the Jacobian of the inverse transformation

$$J = \frac{\partial(x_1, x_2, \dots, x_n)}{\partial(y_1, y_2, \dots, y_n)} = \begin{bmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \dots & \frac{\partial x_1}{\partial y_n} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} & \dots & \frac{\partial x_2}{\partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \frac{\partial x_n}{\partial y_2} & \dots & \frac{\partial x_n}{\partial y_n} \end{bmatrix}.$$

is nonzero in the range of the transformation.

Then $(Y_1, Y_2, \dots, Y_n)$ has a joint absolutely continuous CDF with PDF given by

$$w(y_1, y_2, \dots, y_n) = |J| f(h_1(y_1, \dots, y_n), \dots, h_n(y_1, \dots, y_n)). \quad (1)$$